

Course Project: CryptoPulse

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Soft Deadline 28/06/2026, 06:37 GMT+6

Hard Deadline 28/06/2026, 06:37 GMT+6 100%

The results will be available only after the hard deadline

Project: CryptoPulse

Real-Time Cryptocurrency Analytics Dashboard

Overview

CryptoPulse is a modern frontend application that provides live, real-time analytics for popular cryptocurrencies using public APIs and WebSocket connections. Designed as a responsive, user-friendly dashboard, it displays live price updates, interactive charts, market data, and visual analytics in a single-page interface.

Tech Stack

- React + TypeScript
- WebSockets (Kraken)
- REST API for historical data (Kraken/CoinCap.io)
- Recharts / Chart.js for visualization
- Tailwind CSS or styled-components for styling
- Vite for modern build performance

Core Features

Live Price Updates

Real-time pricing for assets like Bitcoin, Ethereum, Solana, etc., via WebSocket streaming.

Interactive Dashboard

Clean UI with cards displaying current price, percentage change, volume, and market cap.

Dynamic Charts

Display 24h/7d price trends using line charts and sparkline visuals.

Asset List with Sorting & Filtering

Search and sort cryptocurrencies by name, price, market cap, or performance.

Theme Toggle (Dark/Light)

Built with context and local state for user-preferred themes.

Responsive Design

Mobile-friendly layout with flexible components.

Data Source

WebSocket

https://docs.kraken.com/api/docs/websocket-v2/add_order

REST

<https://api.coincap.io/v2/assets>

<https://api.coincap.io/v2/assets/:id/history>

Enhancements

- Real-time order book visualization
- Notifications on large price swings

Estimates

1. Live Crypto Dashboard (WebSocket)

6–8 hours

- Reusable tile component for each asset
- Dynamic styling on price up/down
- Real-time data via https://docs.kraken.com/api/docs/websocket-v2/add_order
- Highlight last change with animation or color

2. Historical Price Charts (REST API)

6–8 hours

- Timeframe toggles (24h, 7d, 30d)
- Line chart with hoverable tooltips
- Uses /history endpoint

- Caching previously loaded data in memory

3. Search, Filter & Sort

5–6 hours

- Search by name or symbol
- Sort by price, market cap, volume
- Filtering by market type (DeFi, Layer 1, etc.)

4. Favorites (Client-Side)

4–5 hours

- Add/remove asset to “favorites”
- Persist to localStorage
- Separate “Favorites” tab or filter

5. Responsive Design & Layout System

5–6 hours

- Mobile & tablet layouts
- Flex/grid layout with adaptive resizing
- Collapsible sidebar, floating header

6. Dark/Light Theme Toggle (via Context)

2–3 hours

- Theme context with toggle switch
- Tailwind/dynamic className or styled-components theme
- Persist preference in localStorage

7. Modular Custom Hooks

3–4 hours

- useWebSocketPrices
- useFetchAssetData
- useTheme

- Each with clean TS typings & error handling

8. Animations & Visual Feedback

2–3 hours

- Animate number updates
- Fade-in cards on load
- Subtle transitions for filter/sort

9. Error States & Skeletons

2–3 hours

- Loading skeletons for tiles & charts
- Retry buttons or error messages
- Graceful fallbacks for API failures

Optional Enhancements

- i18n with English/Russian support
- Unit testing with React Testing Library
- Route protection (mock-only)
- Accessibility improvements

Useful links

APIs

- <https://docs.kraken.com/api/docs/guides/global-intro>
- <https://docs.coinapi.io/market-data/websocket-ds/general>
- <https://pro.coincap.io/api-docs>

Examples

- <https://www.livecoinwatch.com/>
- <https://cointracking.info/dashboard.php?language=en>

Charts

- <https://recharts.org/en-US/examples/CustomContentOfTooltip>

- <https://www.chartjs.org/docs/latest/samples/information.html>

Submission Guidelines

You should submit a link to your GitHub repository with the implemented project.

The repository should include:

- Instructions on how to run the project locally.
- A link to the deployed and working version of the project.
- A link to a short video demo showing how the project works.

The deployed version should demonstrate the main functionality described in the project requirements.